MicroBioAgent: Closed-Loop Planning and Asynchronous Execution for Autonomous Microbiome Analysis

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Abstract

Microbiome data analysis demands customized workflows tailored to specific biological questions, sequencing modalities, and data properties. However, constructing and executing these workflows requires navigating heterogeneous software dependencies, reference databases, and multi-step parameter tuning. Here, we present MicroBioAgent, an autonomous, domain-specific agent framework that translates high-level biological queries into executable, study-specific pipelines across 16S rRNA amplicon, reference-based shotgun, and de novo assembly-based metagenomics. Rather than relying on static pipeline templates, MicroBioAgent employs a closed-loop execution architecture: it dynamically selects and runs bioinformatics tools within user-controlled environments while iteratively inspecting intermediate outputs to adapt downstream analytical decisions. Furthermore, to address the long-running nature of metagenomic computation, we introduce an asynchronous background task-management mechanism that reduces large language model (LLM) token consumption by over 50% while enabling real-time plan adjustment and automated workflow resumption. We systematically evaluated MicroBioAgent across three benchmarks of increasing complexity: (1) paired amplicon and shotgun profiling of tomato rhizosphere microbiomes, (2) taxonomic and functional profiling coupled with machine-learning classification in a colorectal cancer cohort, and (3) de novo metagenomic assembly, metagenome-assembled genome (MAG) binning, and metabolic pathway reconstruction in a polyethylene degradation study. Across all benchmarks, MicroBioAgent faithfully recapitulated the primary biological findings of the original studies, exhibiting high analytical concordance despite expected tool- and database-dependent variations. In systematic robustness evaluations across 27 independent runs spanning three backbone LLMs and three prompt-complexity tiers, MicroBioAgent achieved a 98.8% task completion rate and yielded highly reproducible diversity and differential-abundance estimates. MicroBioAgent establishes a scalable, adaptive paradigm for end-to-end computational microbiome research.

Keywords: microbiome analysis, 16S rRNA, shotgun metagenomics, MAGs, AI agent, asynchronous execution

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1 Introduction

Microbiome studies connect sequencing-data processing with taxonomic and functional characterization and downstream statistics, requiring researchers to select tools and configure software and reference resources [1]. Advances in high-throughput sequencing have enabled comprehensive profiling of microbial communities across plant, human, and environmental ecosystems. However, converting raw sequencing reads into reliable scientific findings demands multi-stage analytical workflows. Researchers must select appropriate computational algorithms, maintain complex software dependencies across isolated package environments, configure massive reference databases, write custom post-processing scripts, and execute downstream statistical tests. These multifaceted technical and bioinformatic hurdles present substantial barriers for life scientists without specialized computational training, often limiting their ability to explore alternate analytical strategies or conduct customized investigations independently [2, 3].

To automate complex analytical pipelines, computational workflow managers and general biomedical AI agents have emerged as two key paradigms. Computational workflow systems such as Snakemake and Nextflow offer reproducible, scalable dependency tracking and parallel execution [4, 5]. Nevertheless, these engines require explicit, predefined execution graphs and pipeline specifications; researchers must still determine which analyses to run, manually structure workflow scripts, and interpret intermediate findings. Concurrently, LLM agents have demonstrated substantial promise in general biomedical problem solving and automated code execution, as seen in AutoBA, Biomni, BioMedAgent, BioMaster, and M3A [6–10].

In the microbiome domain, MetaClaw recently introduced an agent framework that bridges predefined cloud-based upstream workflows (hosted on FlowHub) with customizable local downstream skills [11]. However, this architecture presents two key limitations in practical research. First, relying on external cloud platforms restricts analyses subject to data privacy regulations, transfer bandwidth constraints, or institutional policies that require raw sequencing reads to remain on local infrastructure. Second, MetaClaw monitors long-running computations through repeated polling within the active conversation. For metagenomic tasks requiring hours of processing, continuous status checks rapidly consume conversational context, incur unnecessary token overhead, and prevent researchers from providing feedback or adjusting downstream plans while computations are underway.

To address these challenges, we present MicroBioAgent, a domain-specific agent framework for end-to-end computational microbiome research. MicroBioAgent translates high-level biological objectives into executable analyses without relying on static workflow templates. By pairing modular domain skills with local software environments and reference registries, the agent dynamically selects tools, adjusts parameters, and inspects intermediate outputs to guide downstream steps. To manage compute-intensive workflows efficiently, we introduce an background task coordination mechanism. This mechanism decouples long-running shell execution from the conversational dialogue, eliminating polling overhead while enabling researchers to interactively refine pending analyses before automated workflow resumption.

We systematically evaluate MicroBioAgent across 230 sequencing libraries from three published datasets covering paired amplicon and shotgun profiling, functional metagenomic classification, and long-read genome reconstruction. Across all benchmarks, MicroBioAgent accurately executed the required analytical pipelines and recapitulated the biological findings of the original studies. In addition, systematic robustness evaluations across 27 independent sessions demonstrated high task completion rates and reproducible quantitative outputs, while ablation experiments confirmed that our asynchronous coordination mechanism reduces token consumption by over 50% compared to standard polling baselines.

2 Methods

MicroBioAgent is an autonomous conversational agent designed for end-to-end computational microbiome research, implemented on the open-source OpenClaw runtime (Figure 1). The framework coordinates three core functional components: (1) a modular library of domain-specific skills paired with host runtime and reference registries; (2) a closed-loop execution engine that dynamically plans, executes, and validates multi-step analyses on local infrastructure; and (3) an task management mechanism that decouples long-running bioinformatics computations from the conversational dialogue. By executing shell commands and custom scripts directly on the host system, MicroBioAgent eliminates dependencies on external cloud platforms or

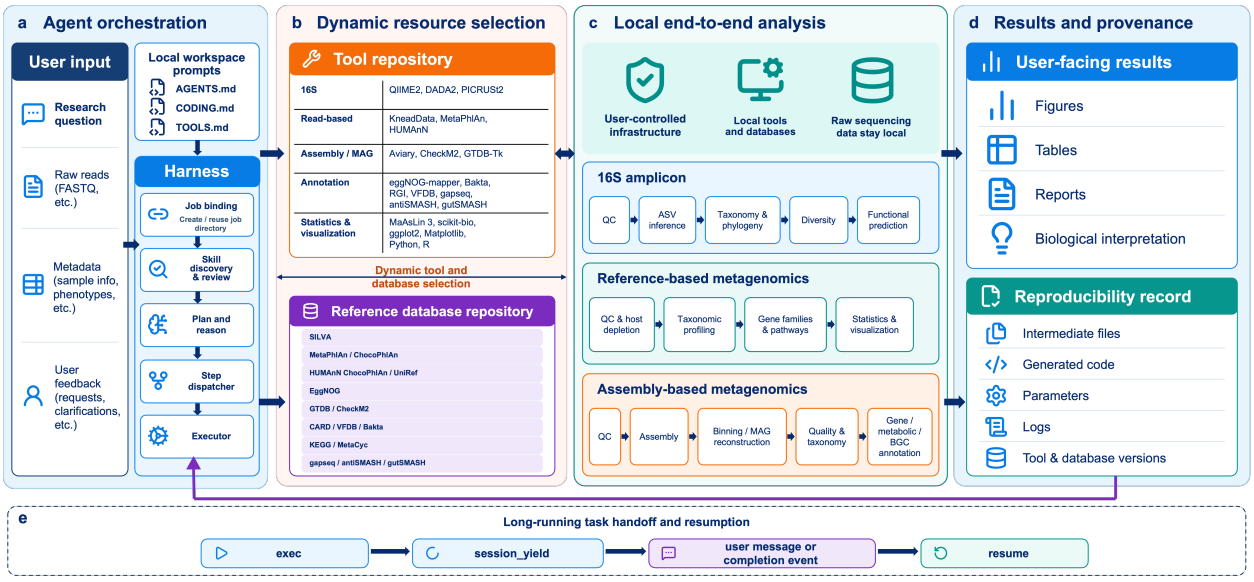


Figure 1: MicroBioAgent architecture and workflow execution. A natural-language research request initiates an analysis workspace, where the agent formulates an analytical plan, selects domain-specific skills, and links local software environments and reference databases across 16S amplicon, reference-based shotgun, and assembly-based metagenomic workflows. Analysis commands, parameters, intermediate tables, and logs are organized within the session directory. During background computation, users can revise pending tasks without interrupting active processes, and task completion triggers output verification and continuation within the same session.

rigid, hardcoded pipeline templates. Implementation details and software/database versions are provided in Appendix A.1 and Table A1.

Analytical domain knowledge is modularized into discrete skills, each defined by a structured `SKILL.md` document specifying its biological objective, required inputs, tool dependencies, reference resources, analytical procedures, and expected deliverables. MicroBioAgent encompasses skills for 16S amplicon processing (QIIME 2, DADA2), shotgun taxonomic profiling (MetaPhlAn), functional quantification (HUMAnN), de novo metagenomic assembly and binning (Aviary, CheckM2), taxonomic placement (GTDB-Tk), coverage calculation (CoverM), functional annotation (eggNOG-mapper), and downstream statistical analysis (MaAsLin 3, DESeq2) [12–22]. Software environments and reference database locations are recorded in two configuration files. `runtimes.yaml` identifies the Conda environment containing each tool or library, whereas `references.yaml` specifies the locations of reference databases such as SILVA, ChocoPhlAn, UniRef, and GTDB [23].

To execute an investigation, the agent first invokes a dedicated workspace tool, `mgx_create_job`, which binds the active conversation to an isolated directory on disk that houses raw inputs, executable scripts, intermediate tables, and logs throughout the session. Within this workspace, MicroBioAgent formulates a structured execution plan based on the research objective, input data, and requested deliverables. The plan defines the sequence of analytical steps and their completion criteria, maintained directly within the conversational context. Guided by workspace instructions (`AGENTS.md`, `TOOLS.md`, `CODING.md`), the agent consults relevant domain skills to author execution scripts and invoke command-line utilities via the `exec` tool. Execution guidelines standardize script authoring to balance compute resources across parallel tasks, ensure comprehensive logging, and verify that commands execute successfully and produce the expected output files. If an execution step fails or expected outputs are missing, the agent inspects the corresponding logs to diagnose the failure and selectively rerun the affected analyses.

To accommodate computationally intensive tasks, MicroBioAgent supports asynchronous background execution with continuous human-in-the-loop interaction. Analytical commands are launched via `exec` with an initial 30-second wait period and no overall execution time limit. Commands that finish within this threshold return their results synchronously. Commands exceeding 30 seconds transition to background processes, prompting the agent to invoke `sessions_yield` to yield its conversational turn. By decoupling

Table 1: Overview of the three case-study datasets.

Case	BioProject	Sequencing libraries	Workflow
Tomato rhizosphere	PRJNA777724	60 16S; 36 matched shotgun	16S and shotgun profiling
CRC faecal metagenomes	PRJEB10878	128 shotgun	Taxonomic and functional profiling
Polyethylene enrichment	PRJEB97175	6 long-read shotgun	Genome-resolved metagenomics

long-running computation from active dialogue, the conversational interface remains responsive, allowing users to inspect intermediate progress, query the agent, or adjust pending analysis plans without interrupting active computations.

To enable automated task resumption after background execution, we implemented a version-pinned runtime extension for OpenClaw (v2026.7.2) that enhances event handling and conversation activation. Each background process is tagged with its originating conversation identifier. When a background task terminates, an event listener captures the exit status, extracts final log records, and routes a structured completion notification to the parent session. If the conversation is idle, the extension immediately initiates a new agent turn without requiring polling or manual user prompts. If an event arrives during an active conversational turn, it is placed in a session-specific queue and delivered once the current turn settles. This design ensures that task results are reliably processed and analysis continues seamlessly upon process completion.

3 Experimental design

3.1 Study-specific analytical capability

Three published datasets comprising 230 sequencing libraries assess paired amplicon and shotgun analysis, colorectal cancer (CRC) functional classification, and long-read genome-resolved metagenomics (Table 1). The tomato rhizosphere dataset (PRJNA777724) evaluates paired amplicon and shotgun profiling across 60 16S (V3–V4) libraries and 36 matched shotgun libraries, spanning four tomato cultivars (two resistant, two susceptible) across baseline and post-inoculation conditions ($n = 3$ biological replicates per group) [24]. The CRC dataset (PRJEB10878) assesses taxonomic and functional profiling across 128 human faecal shotgun metagenomes from 74 CRC patients and 54 controls [25]. The environmental plastic-degradation dataset (PRJEB97175) evaluates de novo genome reconstruction using six Oxford Nanopore long-read metagenomic libraries derived from soil enrichment cultures grown on low-molecular-weight polyethylene (LMWPE) or alkane substrates [26]. In all three cases, the agent was powered by `deepseek-v4-pro` and supplied with raw sequencing reads, sample metadata, and biological objectives without pre-written execution scripts. Evaluation examined analytical task completion and the recovery of primary biological patterns reported in the original studies. Case-study procedures are detailed in Appendix A.2.

3.2 Robustness across models and prompts

To systematically evaluate operational stability while reducing redundant computational overhead, we evaluated three LLMs (`deepseek-v4-pro`, `deepseek-v4-flash`, and `glm5.3-flash`) crossed with three prompt complexity levels (Detailed, Medium, Simple), with three independent sessions per condition ($3 \times 3 \times 3 = 27$ sessions). All runs operated on shared, preprocessed Case 1 data comprising a 16S ASV table, representative sequences, a rooted phylogenetic tree, denoising statistics, sample metadata, and 36 quality-controlled, host-cleared shotgun metagenomic libraries subsampled to 20% read depth. The Detailed prompt explicitly prescribed specific metrics (such as the Shannon index, weighted/unweighted UniFrac, Bray–Curtis, PERMANOVA, and PERMDISP); the Medium prompt defined analytical scope but left tool selection to the agent; the Simple prompt stated only the high-level biological goal [27, 28]. Performance was scored across six core tasks: 16S alpha diversity, 16S beta diversity, 16S genus profiling, 16S differential abun-

dance, shotgun species profiling, and shotgun differential abundance. A task was scored as completed when the corresponding quantitative output was successfully generated. Quantitative consistency was evaluated on aligned subsets using two-sided Mann–Whitney U tests with Holm correction. Complete prompts and additional evaluation procedures are provided in Appendix A.3.

3.3 Ablation of runtime extension

Using `deepseek-v4-pro`, the Detailed Case 1 prompt, and the preprocessed Case 1 inputs, we conducted three independent replicates under the full system (`full`) and three replicates under an ablated configuration lacking the runtime extension (`no-patch`). In the `no-patch` configuration, the system lacked automated completion notifications, and the agent was not guided to release the conversation via `sessions_yield`. All other components—including skills, software environments, and reference databases—were identical. Sessions ran with no user messages following initial plan confirmation. Task completion was scored across nine requested outputs (alpha diversity, beta diversity, 16S genus table, 16S differential abundance, shotgun species table, shotgun differential abundance, figures, execution scripts, and an analytical report), and method adherence was evaluated against the eight diversity and statistical requirements named in the detailed prompt (sequencing-depth normalization, Shannon diversity, weighted UniFrac, unweighted UniFrac, Bray–Curtis dissimilarity, weighted-UniFrac PCoA, PERMANOVA, and PERMDISP). Model calls, tool invocations (`exec`, `process`, `sessions_yield`), and total token consumption (input, output, and reasoning tokens) were recorded from full session transcripts.

3.4 Interactive guidance during background execution

To evaluate human-in-the-loop steering, we tested mid-run plan revision while the Case 1 QIIME 2 taxonomic classification step was running in the background. Once the classification process was confirmed active, a standardized user message instructed the agent to keep the running classification command, switch the subsequent 16S differential-abundance method to DESeq2, and generate a heatmap of significant genera upon completion, with no further user input thereafter. Each session was evaluated on two criteria: (1) whether the agent processed the instruction and updated the pending analysis plan while the background process remained active; and (2) whether the session automatically resumed upon process completion to inspect outputs and execute the revised plan without manual user prompting. Three independent sessions were evaluated under both the `full` and `no-patch` configurations.

4 Results

4.1 Study-specific analytical capability

We first evaluated MicroBioAgent on paired 16S amplicon and shotgun metagenomic datasets from a tomato rhizosphere bacterial-wilt resistance study [24]. Starting from raw reads, the agent executed quality filtering, DADA2 denoising, taxonomic classification against SILVA 138.2, and diversity analyses. Consistent with the original study, pathogen inoculation caused a marked decline in Shannon diversity among susceptible cultivars (A57 and HG70), whereas resistant cultivars (HG64 and HF12) maintained significantly higher diversity post-inoculation at both day 5 ($P = 0.002$) and day 10 ($P = 0.004$; Figure 2). Weighted-UniFrac PCoA separated samples primarily by inoculation status along PCoA 1 (56.3% explained variance), driven by an expansion of pathogenic *Ralstonia* that dominated susceptible plants. Differential-abundance analysis using MaAsLin 3 further identified 6 to 15 genera consistently enriched in resistant cultivars across sampling conditions (Figure A1). In the matched shotgun metagenomes, profiling with MetaPhlAn3 identified enrichment of the *Pseudomonas putida* group in resistant cultivars at baseline (HG64: $P = 0.013$; HF12: $P = 0.009$), but did not detect the second published marker, *Sphingopyxis unclassified*. Prompted by this discrepancy, the researcher instructed the agent to re-analyze the libraries using MetaPhlAn2 (the version used in the original publication) [29]. This guided reanalysis identified both *P. putida* ($P < 0.05$) and *Sphingopyxis unclassified* ($P < 0.1$, matching the source study's threshold) as enriched in resistant cultivars, recapitulating the published taxonomic biomarkers.

In the second benchmark, we evaluated MicroBioAgent on a published colorectal cancer (CRC) cohort comprising 128 human faecal shotgun metagenomes (74 CRC patients and 54 controls) [25] (Figure 3). Starting from raw sequencing reads, the agent executed species-level profiling via MetaPhlAn3 and functional quantification of UniRef90 gene families and KEGG Orthology (KO) groups via HUMAnN3. PCoA of species-level Bray–Curtis dissimilarities showed no distinct visual separation among controls ($n = 54$), early-stage CRC ($n = 36$), and late-stage CRC ($n = 38$), yet PERMANOVA identified significant community differences associated with CRC status ($F = 2.32, P = 0.002$) and disease stage ($F = 1.65, P = 0.009$), whereas clinical covariates (sex, age, BMI) showed no significant effects. Differential-abundance testing identified eight CRC-enriched species and two control-enriched species (Benjamini–Hochberg adjusted $q < 0.05$). Key enriched taxa—*Parvimonas micra*, *Peptostreptococcus stomatis*, *Fusobacterium nucleatum*, and *Solobacterium moorei*—alongside the control-enriched species *Eubacterium ventriosum*, exhibited the same direction of association reported in the original cohort study. A Spearman co-occurrence network constructed from these ten species revealed 16 positive correlations ($|\rho| \geq 0.4, q < 0.05$), with *P. micra* serving as a central hub connected to six other species, including *F. nucleatum* ($\rho = 0.49$) and *S. moorei* ($\rho = 0.53$). In predictive modeling, the agent implemented nested fivefold cross-validation with within-fold mRMR feature selection, training logistic regression classifiers on the top 15 features [30]. The gene-family classifier achieved a pooled out-of-fold AUROC of 0.902 (sensitivity 0.82, specificity 0.89 at Youden's index), substantially outperforming the KO-based classifier (AUROC 0.632, sensitivity 0.41, specificity 0.87), confirming that fine-grained gene-family abundances provided superior diagnostic discrimination.

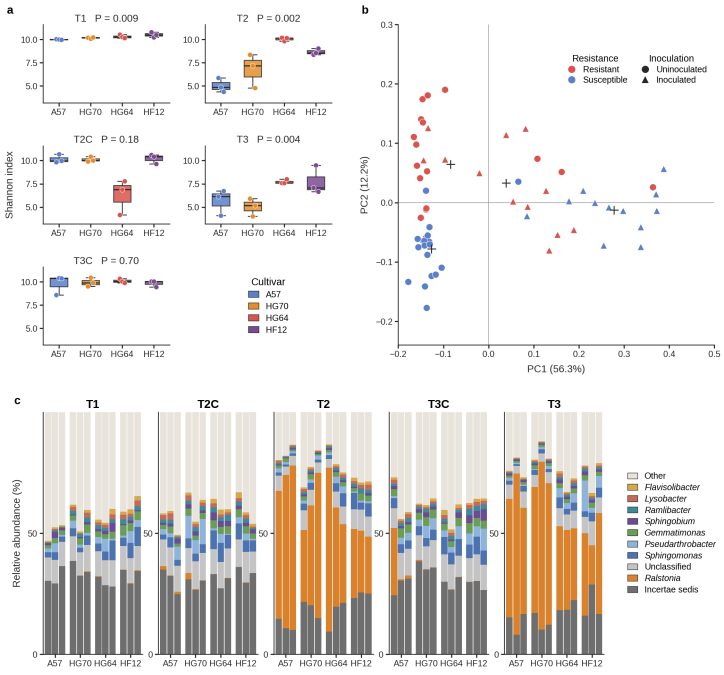


Figure 2: Tomato rhizosphere microbial diversity and community composition. (a) Per-sample Shannon diversity across cultivars, sampling timepoints, and inoculation conditions. (b) PCoA of weighted UniFrac distances, with samples colored by resistance phenotype and point shapes indicating inoculation treatment. (c) Genus-level relative abundance across individual samples. All groups contain $n = 3$ biological replicates ($n = 60$ total 16S samples).

In the third benchmark, we evaluated MicroBioAgent on de novo genome reconstruction and metabolic annotation using an environmental plastic-degradation dataset [26] (Figure 4). The dataset comprised six Oxford Nanopore long-read metagenomes derived from two soil inocula (S1 and S2) enriched on LMWPE, triacontane (C30), or tetracontane (C40). Starting from raw long reads, de novo assembly and binning with Aviary generated 372 initial bins, of which 153 met CheckM2 quality standards (100 high-quality and 53 medium-quality MAGs under MIMAG criteria) [15, 16, 31]. Subsequent dereplication at 95% average

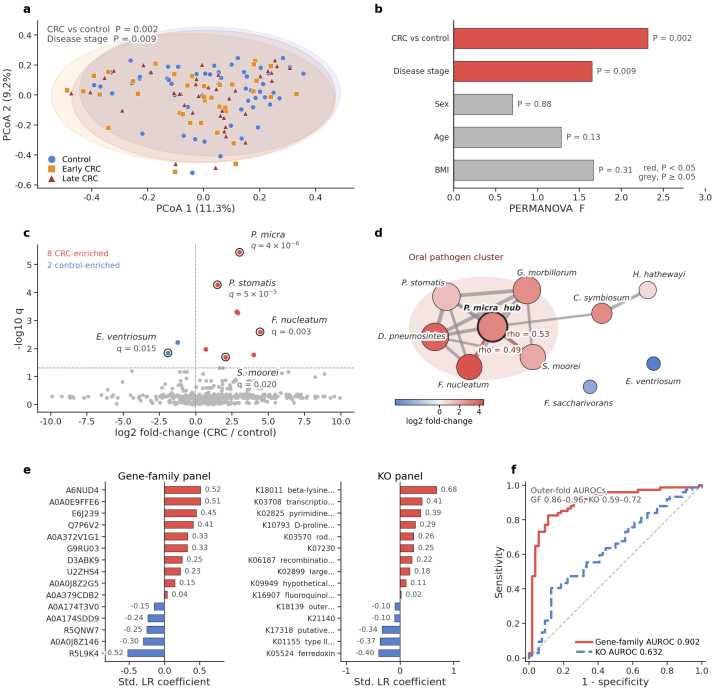


Figure 3: Metagenomic profiling, biomarker discovery, and predictive modeling in the CRC cohort. (a) PCoA of species-level Bray–Curtis dissimilarities across controls, early-stage CRC, and late-stage CRC cases. (b) PERMANOVA test statistics (F values) for disease status, stage, and clinical covariates (sex, age, and BMI). (c) Volcano plot of differentially abundant species between CRC cases and controls (Benjamini–Hochberg adjusted $q < 0.05$). (d) Spearman co-occurrence network of the ten differentially abundant species ($|\rho| \geq 0.4$, $q < 0.05$), highlighting the *P. micra* hub. (e) Standardized logistic regression coefficients for the top 15 gene-family and top 15 KO features selected by mRMR within training folds. (f) ROC curves from nested fivefold cross-validation comparing gene-family and KO classifiers.

nucleotide identity (ANI) yielded 75 non-redundant representative MAGs (50 high-quality and 25 medium-quality). Taxonomic assignment via GTDB-Tk (release R232) and relative abundance estimation via CoverM mapped community structures across carbon sources. In inoculum S1, *Acinetobacter guillouiae* predominated in LMWPE (64%) and C30 (67%) enrichments, whereas *Aquipseudomonas* dominated C40 (66% genus-level abundance). In inoculum S2, *Aquipseudomonas* sp003050835 was most abundant in LMWPE (42%), *Mesorhizobium_M* sp900116595 and *Aquipseudomonas moorei* in C30 (16% each), and *Mesorhizobium_M* sp900116595 and *Aquipseudomonas fluorescens_BJ* in C40 (27% and 25%, respectively). MicroBioAgent recovered all ten focal taxonomic groups reported in the original study (with taxa designated *Pseudomonas* in the original study classified here as *Aquipseudomonas*, consistent with updated GTDB taxonomic boundaries). Functional annotation using eggNOG-mapper across 11 representative MAGs of these focal taxa identified alkane terminal oxidation genes in nine MAGs, while genes for 2-ketone/oxidized-derivative metabolism and the β-oxidation module were detected in all 11 genomes.

4.2 Robustness across models and prompt granularity

To evaluate whether MicroBioAgent maintains reliable execution and consistent analytical outputs across LLMs and prompt granularities, we crossed three LLM backbones (`deepseek-v4-pro`, `deepseek-v4-flash`, and `glm5.3-flash`) with three prompt levels (Detailed, Medium, and Simple) on preprocessed Case 1 inputs ($3 \times 3 \times 3 = 27$ independent sessions) (Figure 5). Across six core analytical tasks, MicroBioAgent achieved an overall task completion rate of 98.8% (160 of 162 evaluations). Both `deepseek-v4-pro` and `deepseek-v4-flash` completed all six analyses in three of three sessions under every prompt level. `glm5.3-flash` completed all 16S analyses (9/9 sessions) and 8 of 9 shotgun sessions, omitting the shotgun workflow in one session under the Detailed prompt. These results demonstrate that the framework's domain

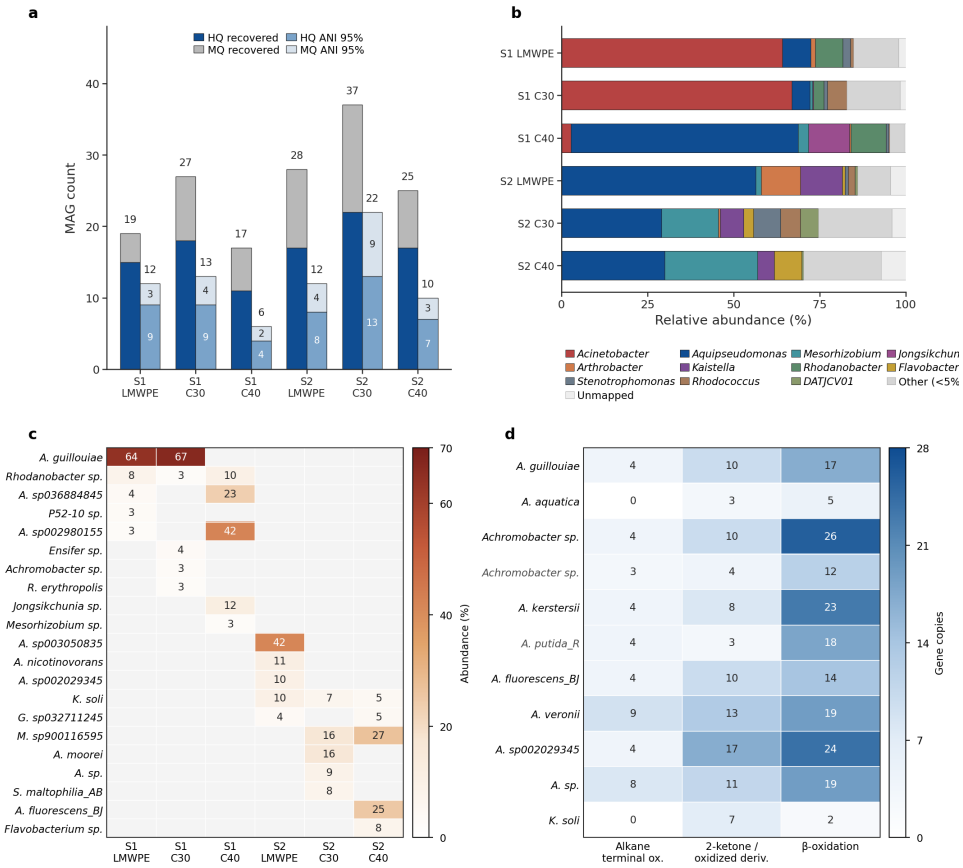


Figure 4: Genome-resolved metagenomic analysis of polyethylene-enrichment cultures. (a) Counts of high-quality and medium-quality MAGs before and after 95% ANI dereplication. (b) Genus-level relative abundance of MAGs across the six enrichment cultures. (c) Relative abundances of species ranked in the top five of any of the six enrichment cultures. (d) Gene copy numbers annotated by eggNOG-mapper in 95% ANI representative MAGs of focal taxa for alkane terminal oxidation, 2-ketone or oxidized-derivative metabolism, and β-oxidation; taxa in gray font denote medium-quality MAGs.

skills and runtime registries provide stable operational guidance across diverse reasoning backbones and varying degrees of prompt specification.

Quantitative outputs for community diversity and differential abundance exhibited broadly concordant distributions across experimental conditions (Figures 5b and A2). Per-sample 16S Shannon diversity indices (median 9.59–9.83) and pairwise weighted UniFrac distances (median 0.301–0.303) showed no statistically significant differences between any pair of models under any prompt level, or across prompt levels within any model (all Holm-adjusted $P = 1.0$, two-sided Mann–Whitney U test). In differential-abundance analyses, individual sessions autonomously chose among DESeq2, MaAsLin 3, Wilcoxon rank-sum, Mann–Whitney U, or LEfSe [32]. Restricting cross-model comparisons to sessions utilizing DESeq2 revealed no significant distributional differences in estimated $\log_2$ fold changes across models (all Holm-adjusted $P \geq 0.059$; Table A2). Similarly, within-model comparisons between Detailed and Medium prompts using matched statistical tests yielded consistent effect-size distributions (all Holm-adjusted $P \geq 0.064$; Table A3).

In contrast, taxonomic relative abundances were more sensitive to model and prompt specifications. Across models, relative-abundance distributions did not differ in 16S analyses under the Simple prompt or in shotgun analyses under the Detailed prompt (all Holm-adjusted $P = 1.0$), but differed in the remaining settings (Holm-adjusted $P \leq 0.05$). Detailed inspection of session transcripts and generated data matrices revealed three recurring procedural causes for these differences: (1) in several 16S amplicon sessions, the agent selected a V4 rather than a V3–V4 classifier during taxonomic assignment; (2) an occasional post-processing

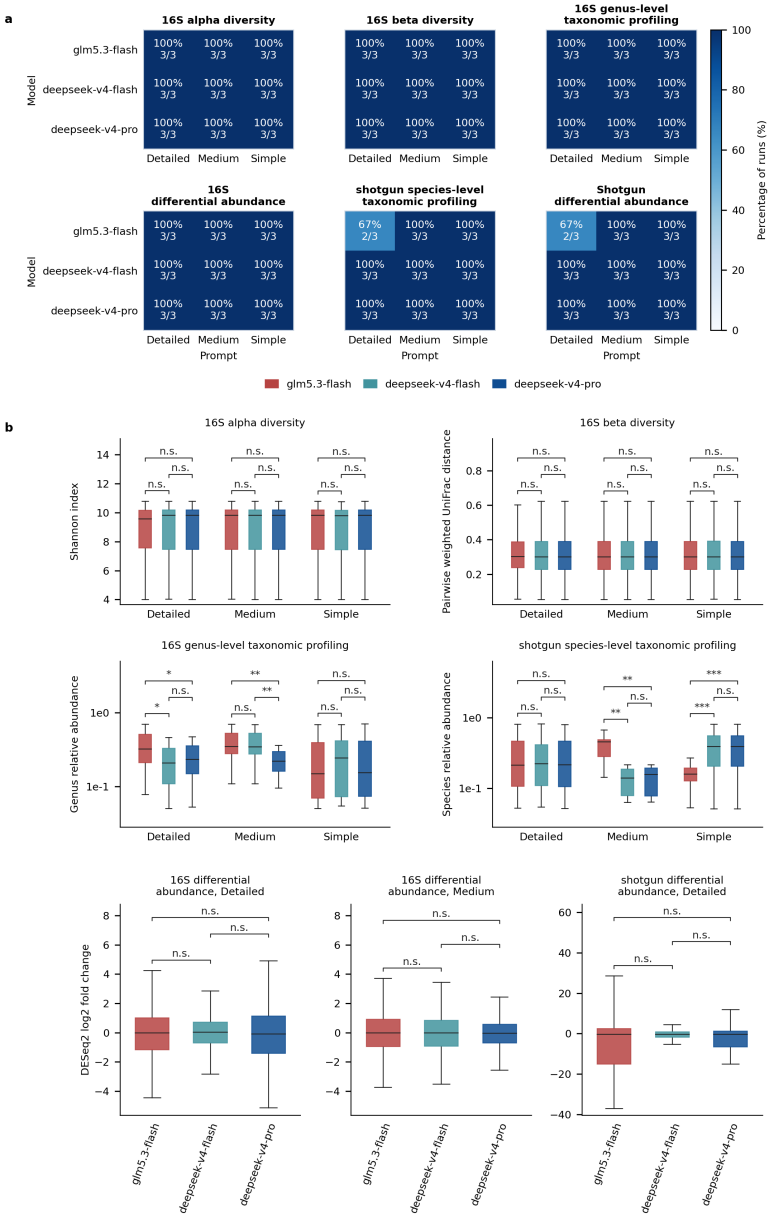


Figure 5: Overall task completion and differences in analytical outputs across models within each prompt level. (a) Completion of six analyses by model and prompt level; values represent completed sessions out of three. (b) Comparison of 16S Shannon diversity indices, pairwise weighted UniFrac distances, and relative abundances (16S genus level and shotgun species level) among models under each prompt level, along with DESeq2 $\log_2$ fold changes in taxon abundance for 16S analyses (detailed and medium prompts) and shotgun analyses (detailed prompt). Brackets indicate two-sided Mann–Whitney U tests with Holm correction across the three model pairs in each plot. n.s., adjusted $P \geq 0.05$; $^{*}P < 0.05$; $^{**}P < 0.01$; $^{***}P < 0.001$. Values are derived from repeated analyses of the same Case 1 dataset and do not represent independent biological replicates.

Table 2: End-to-end execution ablation and asynchronous steering performance ($n = 3$ independent sessions per condition on preprocessed Case 1).

Condition	Outputs	Methods	Model calls	`exec`	`process`	`yield`	Tokens ($\times 10^6$)	Steering / Resumption
`full`	27/27	24/24	133.3	82.3	7.3	11.7	14.53	3/3 / 3/3
`no-patch`	26/27	24/24	286.3	106.0	155.0	0.0	35.08	0/3 / 0/3

script merged sample abundance tables by row position rather than matching feature identifiers; and (3) absent explicit prompt constraints, MetaPhlAn executions differed in whether the `-unknown_estimation` flag was enabled, altering whether species relative abundances were normalized to total reads or classified reads. Overall, MicroBioAgent maintained high task completion and consistent diversity and differential-abundance outputs across models and prompt levels, with remaining differences largely reflecting choices of analytical tools and software parameters.

4.3 Execution efficiency and token consumption

To evaluate how background task coordination affects execution efficiency and resource consumption, we compared the full system (`full`) against the ablated configuration lacking the runtime extension (`no-patch`) across three independent sessions (Table 2). Across all replicates, `full` produced all 27 requested outputs (9 of 9 per session) and satisfied all 24 prespecified methodological requirements (8 of 8 per session). The `no-patch` configuration fulfilled all 24 method requirements and produced 26 of 27 outputs, omitting only the archived analysis scripts in a single session. Overall, omitting the runtime extension did not prevent the agent from completing the requested analyses, but it fundamentally altered how long-running computations were handled.

In `full`, whenever a long-running computational command exceeded the initial 30-second threshold, the agent released conversational control via `sessions_yield` (mean 11.7 calls per session, range 5–17) to await automated completion notifications, rarely polling process status via `process` (mean 7.3 calls, range 0–21, with two sessions making at most one call). In contrast, the ablated `no-patch` configuration never yielded conversational control (0 `sessions_yield` calls) and instead repeatedly polled running tasks, issuing an average of 155.0 `process` queries per session (range 102–227). Because each polling round triggered an active model invocation that reprocessed the expanding conversation transcript, total model calls more than doubled in `no-patch` (mean 286.3, range 211–403) compared to `full` (mean 133.3, range 124–150). Consequently, mean total token consumption rose by 141.5% (a 2.4-fold increase), jumping from 14.53×10^6 tokens in `full` to 35.08×10^6 tokens in `no-patch` (per-session records in Table A4).

4.4 Interactive plan revision during background computation

We next evaluated whether users could interactively revise the pending analysis plan while a minutes-scale QIIME 2 taxonomic classification command was running in the background. Once the classification job was confirmed active, the user sent an instruction to maintain the running process, switch the subsequent differential-abundance method to DESeq2, and generate a heatmap of significant genera upon completion.

In `full`, the agent released conversational control via `sessions_yield` upon starting the task and successfully updated the pending plan in response to the user's instruction while the classification job was actively running (3/3 sessions; Table 2). Once the background process finished, the runtime extension intercepted the completion event and automatically reactivated the session, allowing the agent to verify the taxonomic outputs and continue execution under the revised plan without additional user prompts (3/3). In contrast, in the ablated `no-patch` configuration, the agent did not release the dialogue and instead kept the conversation occupied with repeated `process` calls. Consequently, neither mid-run plan adjustment nor automated workflow resumption upon completion was achieved in any of the replicates (0/3).

5 Discussion

MicroBioAgent demonstrates that modular domain skills, local software environments, and event-driven task management can be unified to translate high-level biological queries into executable microbiome analyses. This framework is supported by three complementary lines of evidence: (1) three diverse case studies establishing flexible end-to-end execution across amplicon, shotgun metagenomic, and genome-resolved workflows; (2) systematic evaluations across 27 independent sessions demonstrating 98.8% task completion and reproducible diversity and differential-abundance estimates across diverse LLMs; and (3) controlled ablations showing that event-driven execution reduces token overhead by over 50% while enabling real-time plan adjustment. Importantly, MicroBioAgent complements traditional workflow managers such as Snakemake and Nextflow rather than replacing them [4, 5]. While workflow engines excel at large-scale, deterministic execution of static pipelines, MicroBioAgent is tailored for exploratory analyses where analytical plans must be adaptively formulated, inspected, and refined in response to intermediate biological findings.

At the same time, our findings highlight that computational execution alone does not guarantee biological validity. The observed procedural discrepancies—such as selecting a V4 rather than a V3–V4 classifier, merging abundance tables by row position rather than matching feature identifiers, and varying the MetaPhlAn `-unknown_estimation` flag—demonstrate that LLM reasoning remains susceptible to subtle bioinformatic pitfalls. In the tomato reanalysis, for example, initial MetaPhlAn3 profiling detected only one of the two reported species markers; the researcher then specified MetaPhlAn2, the version used in the original study, which recovered both. This rerun shows that the recovered markers depended on the software version. Crucially, by preserving executable commands, parameters, and intermediate tables within an inspectable workspace, MicroBioAgent makes these analytical decisions transparent, allowing life scientists to audit each step, diagnose unexpected variations, and steer downstream analyses accordingly.

Several limitations of the current study warrant further investigation. First, our robustness and ablation experiments were evaluated on preprocessed inputs from a single benchmark dataset; future evaluations should examine operational stability across broader cohort scales and raw sequencing modalities. Second, while our ablation quantified the operational benefits of event-driven execution over status polling, the comparison was conducted at the runtime-extension level; finer-grained component-wise ablations and direct comparisons against general-purpose coding agents supplied with identical tools will help isolate the marginal contribution of domain-specific skills. Nevertheless, MicroBioAgent establishes an adaptable, auditable paradigm for autonomous computational research, bridging natural-language biological intent with rigorous, local bioinformatic execution.

6 Conclusion

MicroBioAgent establishes an autonomous, domain-specific framework for end-to-end computational microbiome research. By integrating modular analytical skills, local software environments, and reference registries, the framework translates natural-language biological goals into executable multi-stage workflows without relying on hardcoded pipeline templates. Furthermore, its event-driven task management mechanism decouples long-running bioinformatics computations from conversational dialogue, reducing token overhead by over 50% while enabling seamless, interactive human steering and automated workflow resumption. Across diverse plant, clinical, and environmental datasets, MicroBioAgent demonstrates high analytical robustness and biological concordance, providing a scalable, transparent foundation for autonomous microbiome analysis.

Code and data availability

Code, skills, prompts, environment specifications, and the version-pinned runtime patch are provided through the project repository. `https://github.com/Zeming-LI-Andy/openclaw-microbiome`. Raw sequencing reads are publicly available under BioProject accessions PRJNA777724 (tomato rhizosphere), PRJEB10878 (CRC cohort), and PRJEB97175 (polyethylene enrichment). The LLMs were accessed through Volcano Engine.

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A Supplementary methods and results

A.1 Implementation and reproducibility

The framework is implemented as an extension for OpenClaw. The tool `mgx_create_job` binds each conversation to a dedicated filesystem workspace under a designated root directory. Domain skills under `skills/` provide analytical guidance and, for some steps, reference scripts for QIIME 2, DADA2, MetaPhlAn, HUMAnN, Aviary, CheckM2, GTDB-Tk, CoverM, eggNOG-mapper, and statistical packages including MaAsLin 3 and DESeq2. Computing environments and reference database versions are locked via Conda environment files and centralized YAML registries (`runtimes.yaml` and `references.yaml`). Software and database versions used in the case studies are listed in Table A1.

Table A1: Principal software tools and reference databases used across the three case studies.

Software or reference database	Version or release
QIIME 2	2024.10.1
DADA2	1.30.0; QIIME 2 plugin 2024.10.0
SILVA 138.2	138.2
KneadData	0.12.4; `hg37dec_v0.1`
MetaPhlAn	3.0.7; `mpa_v30_CHOCOPhlAn_201901`
MetaPhlAn	2; `mpa_v20_m200`
HUMAnN	3.0.0.alpha.4; ChocoPhlAn `v296_201901`
MaAsLin3	1.2.0
Aviary	0.13.2; assembler myloasm 0.3.0
CheckM2	1.1.0
CoverM	0.7.0
GTDB-Tk	2.7.2; database R232
eggNOG-mapper	2.1.13; eggNOG 5.0.2

The runtime coordination mechanism is implemented as a version-pinned patch set applied to OpenClaw (v2026.7.2). The extension modifies tool execution handling such that commands executed via `exec` return control after 30 seconds if the process is still running. It registers an asynchronous background process listener that maps active job IDs to originating session IDs. Upon command termination, the listener captures the exit status and dispatches an event notification directly into the originating session's message stream. If the session is currently idle, the notification triggers an immediate agent loop; if the session is currently streaming or processing user input, the notification is queued and delivered as soon as the active turn finishes. In the ablated `no-patch` configuration, this event listener, completion dispatcher, and turn-yielding instructions are removed, and the agent checks completion with `process`.

A.2 Case-study details and extended results

Sequencing reads from PRJNA777724 were processed across four cultivars: susceptible cultivars A57 and HG70, resistant parent HG64, and resistant hybrid HF12. Samples spanned baseline (T1), inoculated day 5 (T2), uninoculated control day 5 (T2C), inoculated day 10 (T3), and uninoculated control day 10 (T3C), with $n = 3$ biological replicates per group. Shotgun profiling with MetaPhlAn3 used database `mpa_v30_CHOCOPhlAn_201901`. When MetaPhlAn3 detected *P. putida* but not *Sphingopyxis unclassified*, the investigator instructed the agent to re-run the profiling with MetaPhlAn2 (`mpa_v20_m200`), which recovered both markers. Genus-level MaAsLin 3 coefficients for HG64 and HF12 are shown in Figure A1.

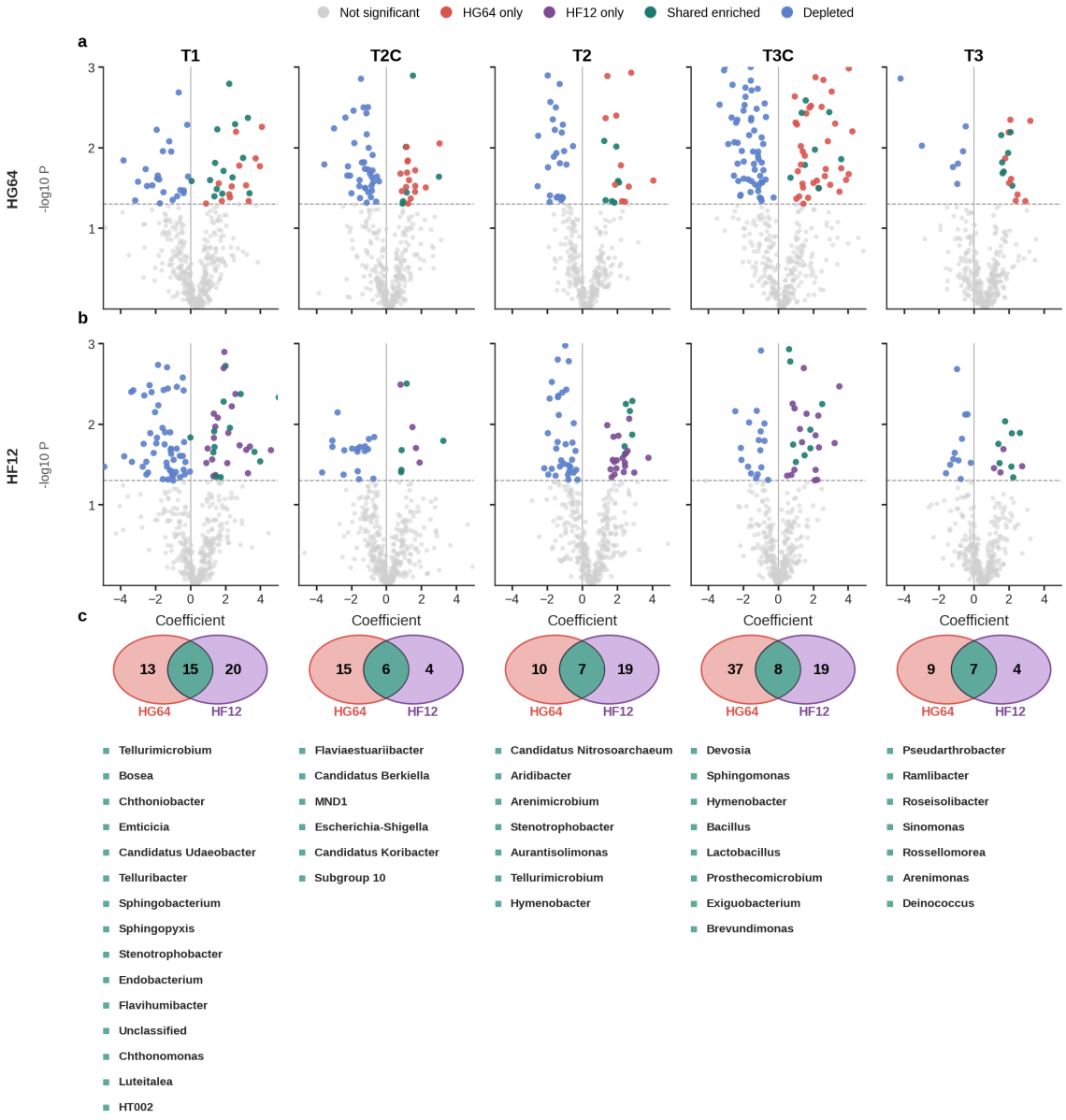


Figure A1: Genus-level differential abundance in the 16S profiles of the resistant tomato parent HG64 and the resistant F1 hybrid HF12. (a, b) Volcano plots of MaAsLin 3 coefficients for HG64 (a) and HF12 (b) versus the pooled susceptible cultivars at each sampling condition. (c) Overlap of genera with a positive coefficient and $P < 0.05$ in both resistant cultivars.

The Hong Kong C1 cohort (PRJEB10878) comprised 128 faecal shotgun metagenomic samples (74 CRC patients and 54 controls). Shotgun reads were profiled with MetaPhlAn3 and HUMAnN3. In predictive modeling, five-fold cross-validation was nested: in each fold, feature selection using minimum redundancy maximum relevance (mRMR) was conducted on the training data to select the top 15 features, followed by logistic regression training. The fitted model was then evaluated on the held-out fold. Out-of-fold predictions pooled across all five folds yielded an AUROC of 0.902 for UniRef90 gene families and 0.632 for KOs.

Oxford Nanopore long-read libraries from PRJEB97175 were assembled and binned using Aviary (incorporating the myloasm assembler), generating 372 raw bins. CheckM2 assessed completeness and contamination, identifying 100 high-quality and 53 medium-quality MAGs. Dereplication at 95% ANI produced 75 non-redundant representative genomes. Taxonomic assignment with GTDB-Tk R232 and coverage quantification via CoverM mapped the abundance of 11 focal MAGs across the six enrichment cultures. eggNOG-mapper annotated alkane terminal oxidation, metabolism of 2-ketones or oxidized derivatives, and β-oxidation.

A.3 Additional evaluation details

The 27 robustness sessions were executed using identical preprocessed Case 1 inputs to eliminate stochastic differences in upstream DADA2 denoising or read trimming. The preprocessed package contained the feature table, representative sequences, rooted phylogenetic tree, and 36 shotgun libraries subsampled to 20% depth. Three prompts (Detailed, Medium, Simple) crossed with three models (`deepseek-v4-pro`, `deepseek-v4-flash`, `glm5.3-flash`) were tested across three replicates. All 162 task evaluations were scored based on the presence of valid quantitative output tables. One session under the Detailed prompt with `glm5.3-flash` omitted shotgun analysis, yielding 160/162 completed evaluations (98.8%). The prompts used in these sessions are reproduced below.

Detailed prompt

You are provided with rhizosphere shotgun metagenomic data and matched 16S rRNA gene amplicon data from a tomato bacterial-wilt experiment. Conduct a comprehensive exploratory and comparative microbiome analysis covering both 16S rRNA and shotgun metagenomic profiling.

Data Characterization and Quality Control

- Describe all available metadata variables and the hierarchical structure of the dataset: cultivar genotype, resistance phenotype, sampling timepoint, inoculation treatment, biological replicate.
- Visualize the experimental design, sequencing depth, and QC retention using sample-flow diagrams and distribution plots stratified by cultivar, timepoint, and treatment.

16S rRNA Alpha Diversity Analysis

- Calculate Shannon diversity for each sample from the post-QC, non-chimeric ASV feature table after sequencing-depth normalization.
- At each sampling timepoint, compare Shannon diversity among cultivars and between resistant and susceptible phenotype groups.
- Visualize Shannon diversity with box/violin plots by cultivar and resistance phenotype, faceted by timepoint and treatment.

16S rRNA Beta Diversity Analysis

- Quantify between-sample dissimilarity from the post-QC ASV feature table using **Weighted UniFrac**, **Unweighted UniFrac**, and **Bray–Curtis**.
- Visualize weighted-UniFrac PCoA with samples colored by phenotype and shaped or faceted by cultivar, timepoint, and treatment.
- Use PERMANOVA to assess resistance phenotype, cultivar, inoculation treatment, sampling timepoint, and prespecified interaction terms. Assess homogeneity of multivariate dispersion with PERMDISP before interpreting PERMANOVA.

Detailed prompt (continued)

16S Taxonomic Profiling and Differential Abundance Analysis

- Conduct taxonomic classification and abundance-composition analysis.
- At each independent sampling timepoint, identify taxa with significantly different abundances between each resistant focal cultivar and the susceptible control group.
- Screen and summarize microbial markers significantly enriched in both resistant cultivars **HG64** and **HF12** at **T1**, **T2C**, and **T2**. Define these shared taxa as core resistance-associated biomarkers.
- Visualize composition with stacked bar plots and differential taxa with volcano plots or heatmaps; summarize shared biomarkers with an UpSet plot.

Shotgun Metagenome Taxonomic and Differential Abundance Analysis

- Perform species-level taxonomic annotation and abundance profiling.
- At each independent sampling timepoint, identify taxa with significantly different abundances between each resistant focal cultivar and the susceptible control group.
- Screen and summarize markers significantly enriched in both **HG64** and **HF12** at **T1**, **T2C**, and **T2** as core resistance-associated biomarkers.
- Visualize species-level composition with stacked bar plots and differential species with volcano plots or heatmaps; summarize shared markers with an UpSet plot.

Medium prompt

You are provided with rhizosphere shotgun metagenomic data and matched 16S rRNA gene amplicon data from a tomato bacterial-wilt experiment. Conduct a comprehensive exploratory and comparative microbiome analysis covering both 16S rRNA and shotgun metagenomic profiling.

Data characterization

- Describe the dataset structure and assess quality.
- Visualize experimental design, sequencing depth, and QC, stratified by cultivar, timepoint, and treatment.

16S diversity, taxonomy, and differential abundance

- Compare alpha and beta diversity among cultivars and between resistant and susceptible groups, stratified by timepoint.
- Profile taxonomic composition.
- At each timepoint, identify taxa that differ between each resistant focal cultivar and the susceptible control.
- Identify markers enriched in both **HG64** and **HF12** at **T1**, **T2C**, and **T2**.

Shotgun taxonomy and differential abundance

- Profile species-level composition.
- At each timepoint, identify species that differ between each resistant focal cultivar and the susceptible control.
- Identify markers enriched in both **HG64** and **HF12** at **T1**, **T2C**, and **T2**.

Simple prompt

You are provided with rhizosphere shotgun metagenomic data and matched 16S rRNA gene amplicon data from a tomato bacterial-wilt experiment. Analyze the dataset and identify microbiome patterns associated with wilt resistance, using both 16S and shotgun profiles.

For alpha and beta diversity comparisons, per-sample Shannon indices ($n = 60$) and pairwise weighted UniFrac distances ($n = 1{,}770$) were aligned across sessions. Taxonomic abundances were compared on shared sample–taxon entries whose session-mean relative abundance was at least 5% in every group being compared. Differential-abundance comparisons included shared taxa without that threshold and were evaluated exclusively across sessions utilizing identical statistical software (Tables A2 and A3; Figure A2). Procedural failures identified in transcripts included: (1) using a V4 classifier rather than a V3–V4 classifier; (2) misalignment of feature identifiers when merging sample profiles; and (3) omission of the `-unknown_estimation` parameter in MetaPhlAn, altering the denominator of relative abundance calculations.

Table A2: Replicate sessions and shared entries for cross-model DESeq2 comparisons.

Dataset	Prompt	DeepSeek Pro	DeepSeek Flash	GLM Flash	Shared entries
16S rRNA gene, genus	Detailed	1/3	2/3	3/3	1,724
16S rRNA gene, genus	Medium	1/3	2/3	1/3	2,659
Shotgun metagenome, species	Detailed	1/3	2/3	2/3	228

Values for each model indicate the number of independent replicates (out of three) that selected DESeq2. Shared entries are the taxa evaluated across all three models. Shotgun results under the medium prompt were not compared because no single differential-abundance method was selected by all three models.

Table A3: Replicate sessions and shared entries for prompt-level differential-abundance comparisons.

Model	Method	Detailed sessions	Medium sessions	Shared entries
16S rRNA gene, genus				
`deepseek-v4-pro`	DESeq2	1/3	1/3	891
`deepseek-v4-flash`	DESeq2	2/3	2/3	5,108
`glm5.3-flash`	DESeq2	3/3	1/3	3,984
Shotgun metagenome, species				
`deepseek-v4-pro`	Wilcoxon	1/3	1/3	39
`deepseek-v4-flash`	Mann–Whitney U	1/3	1/3	120
`glm5.3-flash`	DESeq2	2/3	1/3	52

Session counts indicate the number of independent replicates (out of three) that selected the specified statistical method under each prompt level. Shared entries are the taxa evaluated across both prompt levels.

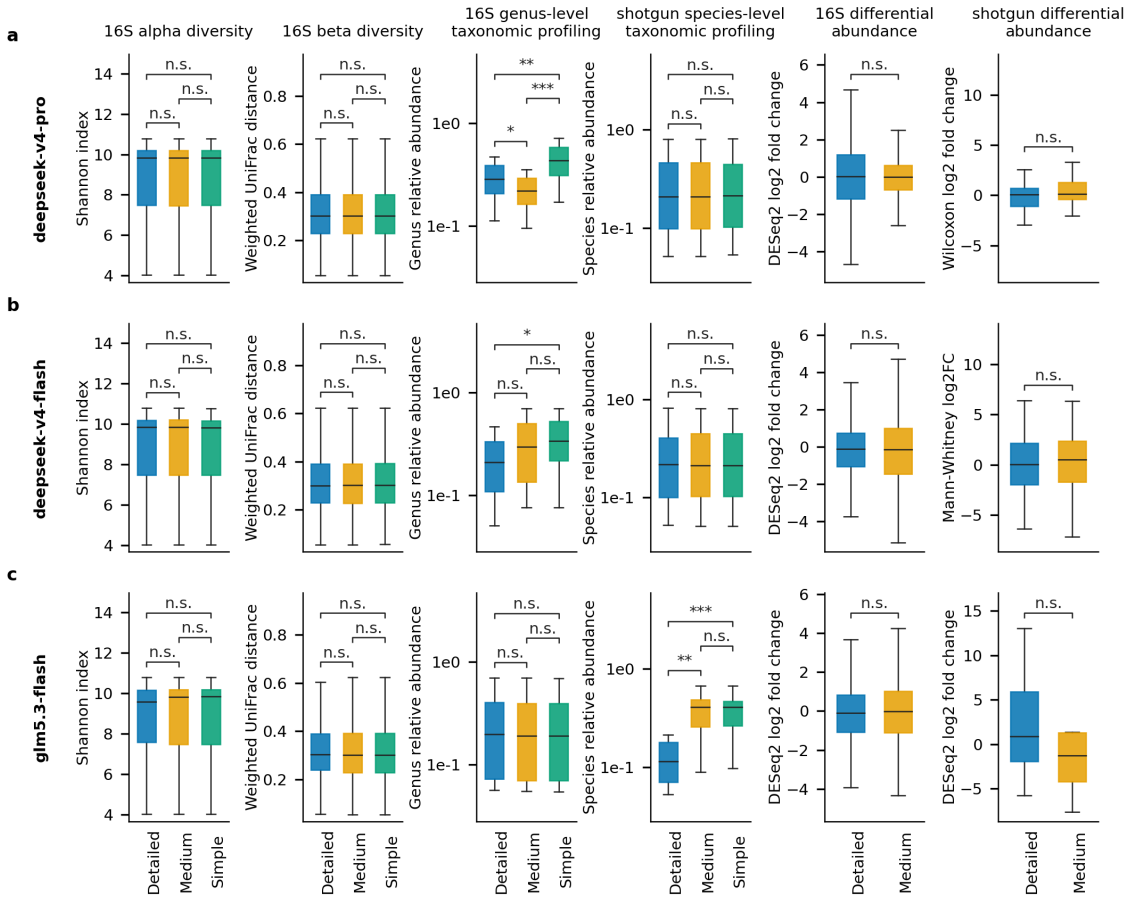


Figure A2: Comparison of analytical outputs across prompt levels within each model. Rows represent `deepseek-v4-pro`, `deepseek-v4-flash`, and `glm5.3-flash`. The first four columns display 16S Shannon diversity indices, pairwise weighted UniFrac distances, 16S genus-level relative abundances, and shotgun species-level relative abundances across prompt levels. The remaining two columns compare differential-abundance $\log_2$ fold changes between the detailed and medium prompts (16S: DESeq2; shotgun: the test shared within each model, Table A3). Brackets in the first four columns indicate two-sided Mann–Whitney U tests with Holm correction across the three prompt pairs within each model; brackets in the differential-abundance panels indicate comparisons between the detailed and medium prompts. $^*P < 0.05$; $^{**}P < 0.01$; $^{***}P < 0.001$; n.s., adjusted $P \geq 0.05$. Values are derived from repeated analyses of the same Case 1 dataset and do not represent independent biological replicates.

Both `full` and `no-patch` used `deepseek-v4-pro`, the Detailed prompt, and the preprocessed Case 1 inputs ($n = 3$ sessions per condition). Across all runs, model calls, tool calls (`exec`, `process`, `sessions_yield`), and token usage (prompt, completion, and reasoning tokens) were recorded. In `full`, the three replicates recorded 150, 124, and 126 model calls, consuming 12.92M, 14.81M, and 15.84M tokens. In `no-patch`, the three replicates recorded 211, 403, and 245 model calls, consuming 23.77M, 49.32M, and 32.15M tokens (Table A4).

Table A4: Per-session outputs, method adherence, tool calls, and token consumption in the execution ablation.

(a) Outputs, method adherence, and model calls

Condition	Session	Outputs (of 9)	Methods (of 8)	Model calls
`full`	1	9/9	8/8	150
`full`	2	9/9	8/8	124
`full`	3	9/9	8/8	126
`full`	Mean / sum	27/27	24/24	133.3
`no-patch`	1	8/9	8/8	211
`no-patch`	2	9/9	8/8	403
`no-patch`	3	9/9	8/8	245
`no-patch`	Mean / sum	26/27	24/24	286.3

(b) Tool calls and token consumption

Condition	Session	`exec`	`process`	`yield`	Tokens ($\times 10^6$)
`full`	1	91	1	17	12.92
`full`	2	74	21	5	14.81
`full`	3	82	0	13	15.84
`full`	Mean / sum	82.3	7.3	11.7	14.53
`no-patch`	1	77	102	0	23.77
`no-patch`	2	142	227	0	49.32
`no-patch`	3	99	136	0	32.15
`no-patch`	Mean / sum	106.0	155.0	0.0	35.08

Summary rows pool outputs and method requirements across the three sessions. Tool-call counts and token consumption in those rows are per-session means. Total tokens are input plus output tokens, including reasoning tokens.

In the human-in-the-loop experiment, the user sent a standardized steering command while QIIME 2 taxonomy classification was executing in the background. In all three `full` sessions, the agent received the message, updated the pending plan to adopt DESeq2 and add a heatmap, yielded its turn, and automatically resumed upon receiving the process completion event. In all three `no-patch` sessions, the agent continued calling `process`; the mid-run instruction was not applied, and the session did not resume automatically.